

Table 1: A comparison of the best Big Data frameworks

	Features	Hadoop	Spark	Flink	Storm	Samza
1	Data processing	It is a batch processing system.	It is a batch processing and stream processing system.	It is a batch processing as well as stream processing system.	It supports stream processing.	It supports stream processing.
2	Streaming engine	MapReduce is a batch-oriented processing tool. It takes large data sets in the input, processes it and produces the results.	Apache Spark streaming processes data streams in micro-batches.	Apache Flink is the true streaming engine. It uses streams for workloads: SQL, micro-batch, and batch.	Storm processes events one by one.	Samza processes events one by one.
3	Data flow	MapReduce computation data flow does not have any loops. It is a chain of stages.	Spark represents it as a direct acyclic graph (DAG).	It supports controlled cyclic dependency graphs in runtime.	Storm's topology is designed as a directed acyclic graph (DAG) with spouts, bolts and streams used to process data.	Samza relies on Kafka's semantics to define the way that streams are handled.
4	Fault tolerance	MapReduce is highly fault-tolerant. There is no need to restart the application from scratch in case of any failure in Hadoop.	Apache Spark streaming recovers lost work and with no extra code or configuration, it delivers exactly-once semantics out-of-the-box.	The fault tolerance is based on Lightweight Distributed Snapshots.	When workers die, Storm will automatically restart them. If a node dies, the worker will be restarted on another node.	Whenever a machine in the cluster fails, Samza works with YARN to transparently migrate your tasks to another machine.
5	Scalability	MapReduce has incredible scalability potential and has been used in the production of tens of thousands of nodes.	It is highly scalable. We can keep adding n number of nodes in the cluster. A large known Spark cluster is of 8000 nodes.	Apache Flink is also highly scalable. We can keep adding n number of nodes in the cluster; a large known Flink cluster is of thousands of nodes.	Scalable, with parallel calculations that run across a cluster of machines.	Samza is partitioned and distributed at every level. Kafka provides ordered, partitioned, fault tolerant streams. YARN provides a distributed environment for Samza.
6	Language support	It supports Java, C, C++, Ruby, Groovy, Perl and Python.	It supports Java, Scala, Python and R.	It supports Java, Scala, Python and R.	Storm is written in Java and Clojure but has good support for non-JVM languages.	Samza is written in Java and Scala. It also supports JVM languages.
7	Latency	Hadoop has higher latency than both Spark and Flink.	It is relatively faster than Hadoop, hence offers lower latency than Hadoop.	With small efforts in configuration, Apache Flink's data streaming runtime achieves low latency and high throughput.	Focuses on extremely low latency.	Offers low latency performance.
8	Cost	MapReduce can typically run on less expensive hardware than some alternatives since it does not attempt to store everything in memory.	As Spark requires a lot of RAM to run in-memory, increasing it in the cluster gradually increases its cost.	Apache Flink also requires a lot of RAM to run in-memory, so the cost of using it will increase gradually.	Apache Storm also requires a lot of RAM to run in-memory, so the cost of using it will increase gradually.	Apache Samza also requires a lot of RAM to run in-memory, so the cost of using it will increase gradually.
9	Security	It supports Kerberos authentication, which is somewhat painful to manage. But third party vendors have enabled organisations to leverage Active Directory Kerberos and LDAP for authentication.	Apache Spark's security is a bit sparse since it currently only supports authentication via shared secret (password authentication). The security bonus is that if you run Spark on HDFS, it can use HDFS ACLs and file-level permissions. Additionally, Spark can run on YARN to use Kerberos authentication.	There is user-authentication support in Flink via the Hadoop/ Kerberos infrastructure. If you run Flink on YARN, Flink acquires the Kerberos tokens of the user that submits programs, and authenticates itself at YARN, HDFS and HBase with that. Flink's upcoming connector and streaming programs can authenticate themselves as stream brokers via SSL.	Storm cluster authenticates users via Kerberos.	Samza uses Apache Hadoop YARN to provide security.
10	Scheduler	The scheduler in Hadoop becomes the pluggable component. There are two schedulers: Fair Scheduler and Capacity Scheduler. To schedule complex flows, MapReduce needs an external job scheduler like Oozie.	Due to in-memory computation, Spark acts as its own flow scheduler.	Flink can use the YARN scheduler but Flink also has its own scheduler.	Storm now has four kinds of built-in schedulers: Default Scheduler, Isolation Scheduler, Multi-tenant Scheduler and Resource Aware Scheduler.	Samza uses the YARN scheduler.